

“Math is Cool” Championships -- 2025-26

7th/8th Grade

Mental Math Solutions

	Answer	Solution
1	50 [= mean]	What is the mean of the numbers forty, sixty, eighty, and twenty? $(20 + 40 + 60 + 80)/4 = 50$
2	350 [millimeters]	How many millimeters are in zero-point-three-five meters? $0.35 \cdot 1000 = 350$
3	21 [candies]	A bag contains only pink and purple candies. All but twelve of the candies are pink and all but nine of the candies are purple. How many candies are in the bag? If all but 12 are pink, then the rest must be purple, so there are 12 purple. If all but 9 are purple, then the rest must be pink, so there are 9 pink. And $9 + 12 = 21$
4	34 [inches]	In inches, what is the perimeter of a rectangle whose length is ten-point-five inches and whose width is six-point-five inches? $(10.5 + 6.5) \cdot 2 = 34$
5	4 [ways]	How many ways are there to arrange the letters in the word SASS, spelled S-A-S-S? ASSS, SASS, SSAS, SSSA – 4 ways
6	480 [seconds]	Anderson completes a one-mile race in six minutes. In this race, Brody runs at an average rate that is three-fourths as fast as Anderson. In seconds, how long does it take Brody to complete the one-mile race? 6 minutes = 360 seconds and $4/3 \cdot 360 = 480$
7	1 [= 6 th term]	The rule in a certain number sequence is: divide the first term by one to get the second term, divide the second term by two to get the third term, divide the third term by three to get the fourth term, and so on. If the first term in this sequence is one hundred twenty, what is the sixth term? 120, 120, 60, 20, 5, 1
8	1080 [= LCM]	What is the least common multiple of ten, twenty-four, and fifty-four? $10 = 2 \cdot 5$, $24 = 2^3 \cdot 3$ and $54 = 2 \cdot 3^3$, so the least common multiple is $2^3 \cdot 3^3 \cdot 5 = 1080$

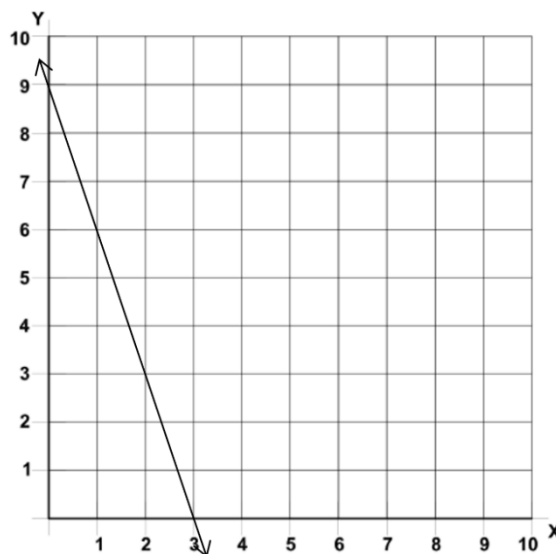
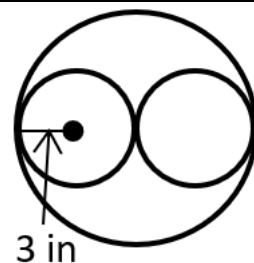
“Math is Cool” Championships -- 2025-26

7th/8th Grade

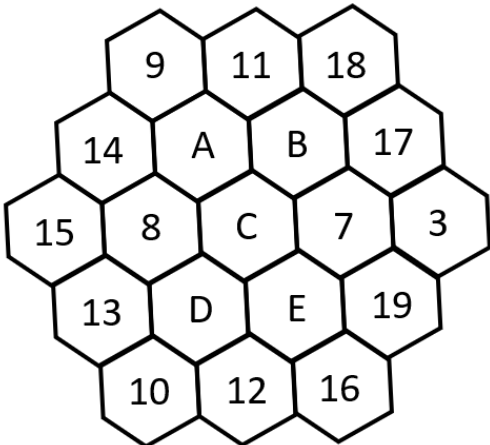
Individual Test Solutions

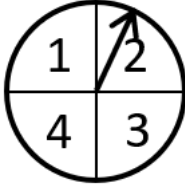
	Answer	Solution
1	822	Evaluate: $901 - 82 + 3$ $901 - 82 + 3 = 822$
2	403 [in ²]	What is the number of square inches in the area of a rectangle that has a width of 13 inches and a length of 31 inches? $13 \cdot 31 = 310 + 93 = 403$
3	40 [minutes]	Billie walks at an average rate of 3 miles per hour. How many minutes does it take her to walk 2 miles? 3 miles in 60 minutes = 2 miles in 40 minutes
4	[A + B =] 20	A jar has 35 green marbles and 30 blue marbles. As a simplified fraction, the probability that a randomly selected marble is green is A/B. What is A + B? $P(\text{green}) = 35/65 = 7/13$ and $7 + 13 = 20$
5	[x =] 7	Solve for x: $29x + 53 = 256$ $29x + 53 = 256 \rightarrow 29x = 203 \rightarrow x = 7$
6	31 [= median]	What is the median of the data set? {31, 28, 33, 35, 26, 39, 30} The median of the set 26, 28, 30, 31, 33, 35, 39 is 31.
7	153 [students]	Ninety percent of the students at Kenroy Elementary School are right-handed. If there are 170 students at Kenroy Elementary, how many are right-handed? $0.9 \cdot 170 = 153$
8	21 [inches]	How many inches are equal to 1.75 feet? $1.75 \cdot 12 = 21$
9	[Q =] 18	A negative integer, P, is 19 units away from a positive integer, Q, on the number line. What is the largest possible value of Q? If P is -1 and Q is 18, then $Q - P = 19$. Q cannot be any bigger without P becoming nonnegative.
10	168 [outfits]	Avril has seven different pairs of pants, six different shirts, and four different pairs of shoes. How many different outfits are available to Avril consisting of one pair of pants, one shirt, and one pair of shoes? $7 \cdot 6 \cdot 4 = 168$
11	34 [= 5 th term]	The first three terms in an arithmetic sequence are -2, 7, and 16. What is the fifth term in the sequence? -2, 7, 16, 25, 34
12	60	What is the sum of the prime numbers between 24 and 32? $29 + 31 = 60$

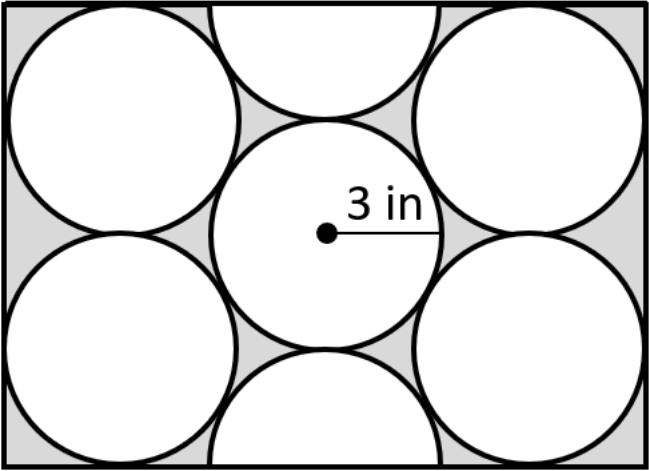
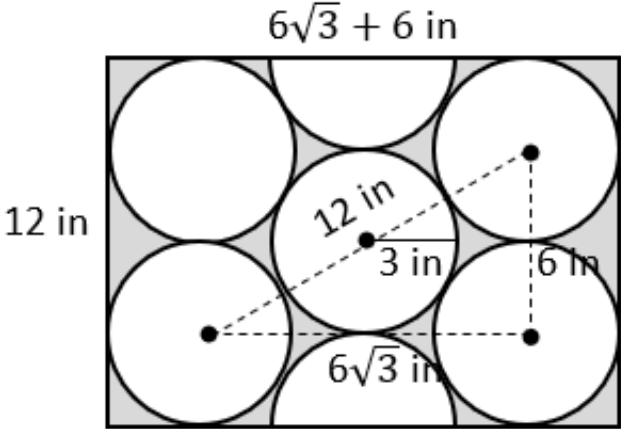
13	972	Evaluate: $12 \cdot 3^4$ $12 \cdot 3^4 = 12 \cdot 81 = 810 + 162 = 972$
14	64 [candies]	A bag contains blue, green, and purple candies. Three-eighths of the candies are blue, $1/4$ of the candies are green, and there are 24 purple candies. How many total candies are in the bag? $3/8 + 1/4 = 3/8 + 2/8 = 5/8$, so 24 purple candies are $3/8$ of the total and 8 purple candies are $1/8$ of the total. Therefore, there are 64 total candies.
15	[P + Q =] 25	The letters A, C, D, and R, are arranged randomly. As a simplified fraction, the probability that they are arranged in the order of the word CARD is P/Q. What is P + Q? There are $4! = 24$ ways to arrange the four letters and 1 of those arrangements is in the order of the word CARD, so the probability is $1/24$, and $1 + 24 = 25$.
16	[A =] 36	Two congruent circles are externally tangent to each other, and both are internally tangent to a third larger circle as shown. The centers of all three circles lie on a diameter of the larger circle. The radius of one of the congruent circles is 3 inches. In terms of π , the area of the large circle is $A\pi$ in ² . What is A? If the radius of the small circle is 3 inches, then the diameter of the small circle is 6 inches, which is the radius of the large circle, so the area of the large circle is $6^2\pi = 36\pi$, so $A = 36$.
17	8 [coins]	Using only quarters (25¢), dimes (10¢), nickels (5¢), and/or pennies (1¢), what is the fewest number of coins needed to make 93¢? Three quarters, 1 dime, 1 nickel, and 3 pennies makes 93¢ and uses 8 coins.
18	9	There are 2 points on the line shown that are in the first quadrant and that have integer coordinates (x, y). What is the sum of the y-coordinates of these 2 points? The points are (1, 6) and (2, 3) and $6 + 3 = 9$.
19	135	The first three terms of a geometric sequence are 5, 15, and 45. What is the fourth term? 5, 15, 45, 135



20	98789	A palindrome is a number that reads the same forwards as backwards, like 101. What is the largest 5-digit palindrome in which the ones, tens, and hundreds digits are all different from each other? The ones digit must be the same as the ten thousands digit and must be the largest it can be or 9. Then the tens digit must be the same as the thousands digit and must be the next largest digit or 8. The hundreds digit must be the next largest digit or 7. So the number is 98789.
21	3 [ways]	How many ways are there to arrange the letters in the word JUNIPER, if the first letter is either E, I, or U, and the remaining letters are in alphabetical order? EIJNPRU, IEJNPRU, UEIJNPR
22	[A + B =] 85	According to the Venn Diagram shown and as a simplified fraction, the probability that a dog-owner also owns a cat is A/B. What is A + B? There are $43 + 21 = 64$ dog-owners and 21 of them own a cat so the probability is $21/64$, and $21 + 64 = 85$.
23	72 [plates]	Animal can wash 24 plates in 10 minutes and Gonzo can wash 18 plates in 15 minutes. Working together, how many plates can Animal and Gonzo wash in 20 minutes? Animal: 24 plates in 10 min = 48 plates in 20 min Gonzo: 18 plates in 15 min = 24 plates in 20 min Together: 72 plates in 20 min
24	3780 [= 6 th term]	An infinite sequence follows the rule that the 2nd term equals the 1st term times 1, the 3rd term equals the 2nd term times 3, the 4th term equals the 3rd term times 5, and so on multiplying each successive term by the next consecutive odd integer to get the following term. If the first term is 4, what is the 6th term? 4, 4, 12, 60, 420, 3780
25	[A + B =] 29	Let $a \nless b = \frac{(a^2+b)}{(b^2-a)}$. Then, as a simplified fraction $12 \nless 6 = A/B$. What is A + B? $12 \nless 6 = \frac{(12^2+6)}{(6^2-12)} = 150/24 = 25/4$, and $25 + 4 = 29$.
26	143[= sum]	The vertices of a triangle are each labeled with a distinct odd, single-digit positive integer. The sides of the triangle are each labeled with the product of the integers at their endpoints. What is the largest possible sum of the labels of the three sides? The largest possible sum is if the vertices are labeled with 5, 7, and 9, which results in the labels of the sides being 35, 45, and 63, and $35 + 45 + 63 = 143$.
27	[A + B =] 7	Emery eats $1/4$ and Liam eats $1/6$ of a cake. Then Vienna eats $1/7$ of what remains. Then Zeke eats $2/3$ of what remains after Vienna eats her portion. Then Ariel eats the rest. As a simplified fraction, Ariel eats A/B of the cake. What is A + B? $1 - 1/4 - 1/6 = 7/12 \rightarrow 7/12 - 1/7 \cdot 7/12 = 1/2 \rightarrow 1/2 - 2/3 \cdot 1/2 = 1/6$, and $1 + 6 = 7$

28	[R =] 38	A set of seven distinct positive integers has a mean of 12, which is also the median. What is the largest possible range, R, where R = the largest integer minus the smallest integer in the set. The seven integers must add up to $7 \cdot 12 = 84$, and the middle number is 12. The set with the maximum range is 1, 2, 3, 12, 13, 14, 39, so $R = 39 - 1 = 38$.
29	17 [cm]	A rectangle has integer side lengths and an integer diagonal length. If the area of the rectangle is 120 cm^2, what is the number of centimeters in the length of the diagonal? Length and Width (L, W) can be (1, 120), (2, 60), (3, 40), (4, 30), (5, 24), (6, 20), (8, 15) or (10, 12) or the reverse of these ordered pairs. The only ordered pairs that can be the legs of a right triangle whose hypotenuse is an integer is (8, 15) or (15, 8), in which case the hypotenuse is 17, which is also the diagonal of the rectangle.
30	[x + y + z =] 13	The expression $\frac{(a^5b^7c^9)^2(a^3b^4c^7)^4}{ab^2c^4}$ can be written as $(a^x b^y c^z)^7$. What is the value of $x + y + z$? $\frac{(a^5b^7c^9)^2(a^3b^4c^7)^4}{ab^2c^4} = \frac{(a^{10}b^{14}c^{18})(a^{12}b^{16}c^{28})}{ab^2c^4} = \frac{a^{22}b^{30}c^{46}}{ab^2c^4} = a^{21}b^{28}c^{42} = (a^3b^4c^6)^7$, and $x + y + z = 3 + 4 + 6 = 13$.
31	12 [= sum]	The five-digit integer 49A70 is divisible by 30, where A represents a non-negative single-digit integer. What is the sum of all possible values of A? For 49A70 to be divisible by 30 it also must be divisible by 2, 3, and 5. Since the ones digit is 0 it must be divisible by 2 and 5. To be divisible by 3, the digits must add up to a multiple of 3, so A could be 1, 4, or 7, and $1 + 4 + 7 = 12$.
32	61542 [=ABCDE]	In the Magic Hexagon shown, the consecutive integers 1 through 19 are distributed such that the sum of all 5 horizontal rows, and of both sets of 5 diagonal rows (the ones sloping up to the right and the ones sloping down to the right) is 38. There is exactly one set of 5 distinct integers that can replace the letters A, B, C, D, and E to ensure the sum of 38 in all rows and diagonals. These integers when listed in order ABCDE create a 5-digit integer. What is this 5-digit integer? The 5 missing integers are 1, 2, 4, 5, and 6 $A = 38 - 11 - 8 - 13 = 6$ $B = 38 - 11 - 7 - 19 = 1$ $C = 38 - 15 - 8 - 7 - 3 = 5$ $D = 38 - 14 - 8 - 12 = 4$ $E = 38 - 17 - 7 - 12 = 2$ The final answer is 61542. 

33	444 [= mean]	The digits 3 and 5 are used to create 8 distinct three-digit positive integers. These three-digit integers can consist of all 3s, all 5s, or a combination of 3s and 5s. What is the mean of these 8 integers? The integers are: 333, 533, 353, 335, 355, 535, 553, 555 $(333 + 533 + 353 + 335 + 355 + 535 + 553 + 555)/8 = (3200 + 320 + 32)/8 = 444$
34	[A + B =] 79	The spinner shown, which is divided into 4 equal sections, is spun 5 times. As a simplified fraction, the probability that all four numbers are spun at least once is A/B. What is A + B?  Total possible outcomes = $4^5 = 1024$ For two 1s, the list begins with the 10 permutations 11234, 21134, 23114, 23411, 12134, 21314, 23141, 12314, 21341, 12341, and the order of the 2, 3, and 4 can be arranged $3! = 6$ ways, so $6 \cdot 10 = 60$. There would also be 60 ways each to arrange the 5 numbers when all 4 numbers are present with two 2s, two 3s, and two 4s, so $4 \cdot 60 = 240$ and $240/1024 = 30/128 = 15/64$ and $15 + 64 = 79$
35	[A + B =] 62	Alek and Aurelia are walking due east side-by-side at the same rate. Alek steps onto an eastward moving walkway and continues walking due east at the same rate as before, while Aurelia increases her rate by 24%. When Alek reaches the end of the moving walkway, Aurelia is 42% of the way to the end of the walkway. As a simplified fraction, the ratio of Alek and Aurelia's original walking rate to the rate of the walkway is A/B. What is A + B? Let Alek and Aurelia's original walking rate be P meters/second. Let the rate of the walkway be Q meters/second. The distance Alek travels while on the walkway is given by the equation $d = (P + Q)t$ and the distance Aurelia travels while Alek is on the walkway is $0.42d = 1.24Pt \rightarrow d = 1.24Pt/0.42$. Then $(P + Q)t = 1.24Pt/0.42 \rightarrow 0.42(P + Q) = 1.24P \rightarrow 0.42P + 0.42Q = 1.24P \rightarrow 0.42Q = 0.82P \rightarrow P/Q = 0.42/0.82 = 42/82 = 21/41$, and $21 + 41 = 62$.
36	[A + B =] 8	The graph of $y = -x + 2$ intersects with the graph of the parabola $y = \frac{1}{2}x^2 - 2$ in two points (x, y). The distance between these two points is $A\sqrt{B}$ units, where A and B are single-digit integers and B has no perfect square factors other than 1. What is A + B? $x + 2 = \frac{1}{2}x^2 - 2 \rightarrow -2x + 4 = x^2 - 4 \rightarrow x^2 + 2x - 8 = 0 \rightarrow (x - 2)(x + 4) \rightarrow x = 2$ and $x = -4$, so the points of intersection are (2, 0) and (-4, 6) and the distance between them is $\sqrt{(2 - (-4))^2 + (0 - 6)^2} = \sqrt{6^2 + 6^2} = \sqrt{72} = 6\sqrt{2}$ and $6 + 2 = 8$.
37	7 [ordered pairs]	How many ordered pairs of positive integers satisfy the following equation? $\frac{5}{a} - \frac{6}{b} = \frac{1}{2}$ The value of b cannot be a positive integer for a-values larger than 9, because for values larger than 9, $5/a \leq 1/2$ and you cannot subtract something positive from a number that is less than or equal to $1/2$ to get $1/2$. Check the b-value for a = 1 through 9. (a, b) = (1, 4/3), (2, 3), (3, 36/7), (4, 8), (5, 12), (6, 18), (7, 28), (8, 48), (9, 108) – answer is 7
38	80 [= 33 rd term]	In a geometric sequence of positive numbers, if the 21st term is 500 and the 39th term is 32, what is the 33rd term? Let r be the common ratio of the sequence, then $500 \cdot r^{18} = 32 \rightarrow r^{18} = 32/500 = 8/125 \rightarrow (r^{18})^{(1/3)} = (8/125)^{(1/3)} \rightarrow r^6 = 2/5 \rightarrow (r^6)^2 = (2/5)^2 \rightarrow r^{12} = 4/25$. The 33rd term will be $500 \cdot r^{12} = 500 \cdot 4/25 = 80$

39	$[A + B + C =] 198$	In the figure shown below there are five complete circles and two partial circles all with the same radius that are externally tangent to each other and internally tangent to a rectangle. Expressed in terms of π, the area of the shaded region is $A\sqrt{3} + B - C\pi \text{ in}^2$, where A, B and C are positive integers. What is $A + B + C$?  The triangle that is added to the figure below is a right triangle and it must be a 30-60-90 triangle since the hypotenuse is double the short leg. Since the two partial circles have the same radius as the five complete circles, their centers must lie on the top and bottom edges of the rectangle, which means they are half-circles, so together they make a 6th circle. The area of the shaded region is the area of the rectangle, $12(6\sqrt{3} + 6) = 72\sqrt{3} + 72$, minus the area of the 6 circles, $6 \cdot 9\pi = 54\pi$, or $72\sqrt{3} + 72 - 54\pi$, and $72 + 72 + 54 = 198$. 
40	6 [values]	Let N be an integer whose prime factorization is in the form $a^3 \cdot b^2 \cdot c$, where a, b, and c are distinct positive prime numbers. How many values of N are less than 900? $2^3 \cdot 3^2 \cdot 5 = 360$ $2^3 \cdot 3^2 \cdot 7 = 504$ $2^3 \cdot 3^2 \cdot 11 = 792$ $2^3 \cdot 3^2 \cdot 13 > 900$ $2^3 \cdot 5^2 \cdot 3 = 600$ $2^3 \cdot 5^2 \cdot 7 > 900$ $2^3 \cdot 7^2 \cdot 3 > 900$ $3^3 \cdot 2^2 \cdot 5 = 540$ $3^3 \cdot 2^2 \cdot 7 = 756$ $3^3 \cdot 2^2 \cdot 11 > 900$ $3^3 \cdot 5^2 \cdot 2 > 900$ $5^3 \cdot 2^2 \cdot 3 > 900$ There are 6 values less than 900.

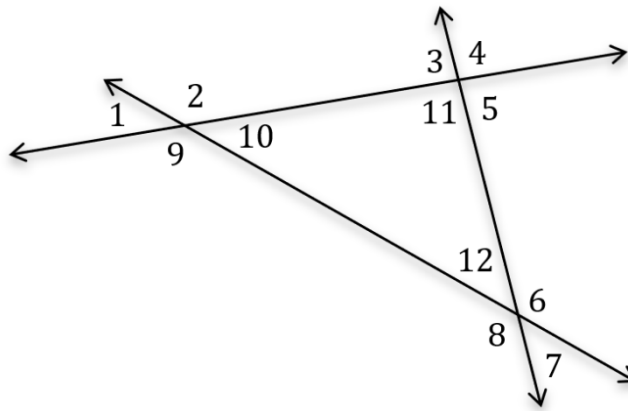
“Math is Cool” Championships -- 2025-26

7th/8th Grade

Multiple Choice Solutions

	Answer	Solution
1	D	Evaluate the expression: $2^1 + 2^2 + 2^3 + 2^4 + 2^5$ A) 26 B) 30 C) 61 D) 62 E) None of the above $2 + 4 + 8 + 16 + 32 = 62$
2	E [5816]	Evaluate the expression: $10^1 - 9^2 + 8^3 - 7^4 + 6^5$ A) 18 B) 1394 C) 4816 D) 6874 E) None of the above $10 - 81 + 512 - 2401 + 7776 = 8298 - 2482 = 5816$
3	B	Evaluate the expression: $25^2 + 15^4 + 5^6 - 23^2 - 13^4 - 3^6$ A) 84 B) 37056 C) 38336 D) 39616 E) None of the above $25^2 + 15^4 + 5^6 - 23^2 - 13^4 - 3^6 = 25^2 - 23^2 + 15^4 - 13^4 + 5^6 - 3^6 = (25 + 23)(25 - 23) + (15^2 + 13^2)(15^2 - 13^2) + (5^3 + 3^3)(5^3 - 3^3) = (48)(2) + (225 + 169)(225 - 169) + (125 + 27)(125 - 27) = 96 + 394 \cdot 56 + 152 \cdot 98 = 96 + (400 - 6)(50 + 6) + (160 - 8)(90 + 8) = 96 + 20000 - 300 + 2400 - 36 + 14400 - 720 + 1280 - 64 = 96 + 38080 - 1120 = 38176 - 1120 = 37056.$ Or use brute force.

REFER TO THE FIGURE SHOWN HERE, CONSISTING OF THREE INTERSECTING LINES, TO SOLVE PROBLEMS #4 THROUGH #6.



4	C	If $m\angle 1 = 38^\circ$ and $m\angle 4 = 98^\circ$, what is $m\angle 12$? A) 34° B) 36° C) 44° D) 54° E) None of the above $m\angle 10 + m\angle 11 + m\angle 12 = 180$ and $m\angle 1 = m\angle 10$ and $m\angle 4 = m\angle 11$, so $m\angle 12 = 180 - m\angle 1 - m\angle 4 = 180 - 38 - 98 = 44$.
5	E	If $m\angle 2 = (17n + 3)^\circ$, $m\angle 3 = (11n - 5)^\circ$, and $m\angle 7 = (4n + 10)^\circ$, then what is $m\angle 6$? Note: this may result in a different set of angle measures from the ones in problem 4. A) 42° B) 98° C) 128° D) 136° E) 138° $(180 - m\angle 2) + (180 - m\angle 2) + m\angle 7 = 180 \rightarrow 180 - (17n + 3) + 180 - (11n - 5) + 4n + 10 = 180 \rightarrow 360 - 17n - 11n - 3 + 5 + 4n + 10 = 180 \rightarrow 372 - 24n = 180 \rightarrow 192 = 24n \rightarrow n = 8$ $m\angle 7 = 4 \cdot 8 + 10 = 42$ and $m\angle 6 = 180 - m\angle 7 = 180 - 42 = 138$.
6	B	If $m\angle 11 > m\angle 12 > m\angle 10$, then the triangle formed by the three intersecting lines in the figure is a triangle in which all three sides are different lengths, i.e. a scalene triangle. If the side that is neither shortest nor longest is 18 cm in length, then how many distinct sets of integer side lengths, measured in centimeters, are possible for this triangle? A) 120 B) 136 C) 144 D) 153 E) infinite (1, 18, 19) is impossible since the length of any given side must be less than the sum of the other 2 side lengths. (2, 18, 19) - 1Δ (3, 18, 19 or 20) - 2Δs (4, 18, 19 - 21) - 3Δs ... (17, 18, 19 - 34) - 16Δs $1 + 2 + \dots + 15 + 16 = 17 \cdot 16/2 = 136$

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #7 THROUGH #10.

An arithmetic sequence is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, 3, 9, 15, 21, 27 is an arithmetic sequence with five terms, in which the first term is 3 and the constant added is 6, also known as the common difference.

A geometric sequence is a sequence in which each term after the first is obtained by multiplying the previous term by a constant. For example, 6, 12, 24, 48, 96 is a geometric sequence with five terms, in which the first term is 6 and the constant that terms are multiplied by is 2, also known as the common ratio.

7	A	The first term of an arithmetic sequence is 23. What is the common difference that results in a fifth term of 41? A) 4.5 B) 5 C) 5.5 D) 6 E) None of the above $41 - 23)/4 = 4.5$																																
8	B	The common ratio of a geometric sequence is $2/3$. What is the first term in the sequence if the second term is 10? A) $20/3$ B) 15 C) 18 D) 20 E) None of the above If x is the first term, then $x \cdot 2/3 = 10 \rightarrow x = 3/2 \cdot 10 = 15$																																
9	D	The first row in this 4×4 array is an arithmetic sequence with 4 terms, and each column is a geometric sequence with 4 terms. Three of the squares are labeled with A, B, and C, as shown. What is $A + B + C$? A) 93 B) 128 C) 137 D) 143 E) 154 The completed array is as shown. <table><tr><td>1</td><td>3</td><td>5</td><td>7</td></tr><tr><td>5</td><td>12</td><td>15</td><td>14</td></tr><tr><td>25</td><td>48</td><td>45</td><td>28</td></tr><tr><td>125</td><td>192</td><td>135</td><td>56</td></tr></table> $A + B + C = 3 + 15 + 125 = 143$ <table><tr><td>1</td><td>A</td><td></td><td>7</td></tr><tr><td>5</td><td></td><td>B</td><td>14</td></tr><tr><td></td><td></td><td>45</td><td></td></tr><tr><td>C</td><td>192</td><td></td><td></td></tr></table>	1	3	5	7	5	12	15	14	25	48	45	28	125	192	135	56	1	A		7	5		B	14			45		C	192		
1	3	5	7																															
5	12	15	14																															
25	48	45	28																															
125	192	135	56																															
1	A		7																															
5		B	14																															
		45																																
C	192																																	

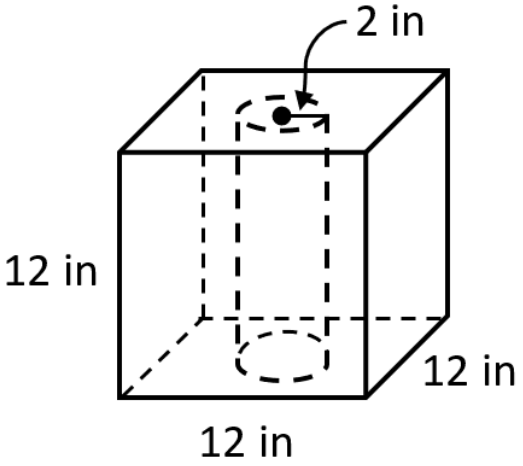
10	D	Consider the sequence of positive integers 1, 2, 1, 3, 6, 3, 1, 2, 5, 8, 5, 2, 1, 3, 6, 9, 12, 9, 6, 3, 1, 2, 5, 8, 11, 14, 11, 8, 5, 2, 1, ... The rule for the sequence is that between the first pair of 1s is a 2, then between the next pair of 1s is 363, then between the next pair of 1s is 25852, and so on. The set of numbers between each successive pair of 1s alternates between an arithmetic sequence starting with 2 and an arithmetic sequence starting with 3, both of which have a common difference of 3 until reaching the middle number in the set, followed by a common difference of -3. What is the 2026th term in this sequence? A) 1 B) 132 C) 133 D) 134 E) 135 There are $1 + 3 + 5 + \dots + n$ terms that are not 1s and $n + 1$ terms that are 1s. For example, for the first six 1s there are $1 + 3 + 5 + 7 + 9 = 25$ terms that are not 1s, and the total number of terms in this example is $25 + 6 = 31$. The sum of the first n odd numbers is n^2. So, the number of 1s, which is $n + 1$, plus the sum of the first n odd numbers, which is n^2, must be less than or equal to 2026 $\rightarrow n^2 + n + 1 \leq 2026$. Using guess and check, the largest value of n is 44 and $44^2 + 44 + 1 = 1981$, so the 1981st term is a 1 followed by the 45th odd number of non-1 terms. The even-ranked odd numbers of terms will start with 3 and the odd-ranked numbers of terms will start with 2, so the 45th odd number is 89, and this set of 89 terms that aren't 1s consists of 2, 5, 8, ..., 134, ..., 8, 5, 2. We want the $2026 - 1981 = 45^{\text{th}}$ term in this group, which is 134.
----	---	--

“Math is Cool” Championships -- 2025-26

7th/8th Grade

Team Test Solutions

	Answer	Solution
1	$[A + B =] 14$	A plastic jug contains 4 red, 6 blue, and 8 purple balls. As a simplified fraction, the probability that a randomly selected ball is not purple is A/B . What is $A + B$? $P(\text{not } P) = 10/18 = 5/9$ and $5 + 9 = 14$
2	1	Evaluate: $2^4 - 6 \cdot \sqrt{25} + \frac{90}{6}$ $2^4 - 6 \cdot \sqrt{25} + 90/6 = 16 - 6 \cdot 5 + 15 = 16 - 30 + 15 = -14 + 15 = 1$
3	48 [minutes]	Bert painted one-third of a fence, and Ernie painted one-fourth of the same fence, each at a constant rate. If Bert, continuing at the same rate, finished painting the rest of the fence in 20 minutes, how many minutes would it have taken him to paint the entire fence by himself? $1 - 1/3 - 1/4 = 5/12$, so the part that took Bert 20 minutes is $5/12$ of the fence. It takes 4 minutes to paint $1/12$ of the fence, so the whole fence would take 48 minutes.
4	13 [multiples]	How many positive 2-digit multiples of 7 are there? 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98 makes 13.
5	28	A data set consists of 8 distinct positive odd integers and has a range of 20. One of the integers is 15. What is the largest possible median for this data set? Let 15 be the smallest number in the set, then 35 is the largest. If the other 6 numbers are 23, 25, 27, 29, 31, and 33, then the median is 28.
6	$[x =] -57$	Solve for x : $11 - 4x = x + 125 - 3x$ $11 - 4x = x + 125 - 3x \rightarrow 11 - 4x = 125 - 2x \rightarrow -2x = 114 \rightarrow x = -57$

7	$[A + B =]$ 904	A cube has a side length of 12 inches. A cylindrical hole with a radius of 2 inches is drilled through the center of the top and the bottom of the cube. In square inches, the surface area of the solid, including the surface of the cylindrical hole, is $A + B\pi$, where A and B are positive integers. What is $A + B$? The surface area of a cube with sides of 12 is $6 \cdot 12^2 = 6 \cdot 144 = 864$. You then have to subtract the two circles, $2 \cdot 2^2\pi = 8\pi$, and add the lateral surface area of the cylinder, $4\pi \cdot 12 = 48\pi$, so the total surface area is $864 - 8\pi + 48\pi = 864 + 40\pi$, and $864 + 40 = 904$. 
8	$[p + a =]$ 32	Consider the sequence of numbers 1, 6, 16, 36, 76, . . . , where each number after 1 is four more than two times the previous number. The positive difference between the 31st and 32nd numbers in the sequence can be written as $5 \cdot p^a$, where p is a prime number and a is an integer. What is the value of $p + a$? For the sequence 1, 6, 16, 36, 76 . . . , the differences can be expressed as: $5 \cdot 2^0$ between the 1st and 2nd terms, $5 \cdot 2^1$ between the 2nd and 3rd terms, $5 \cdot 2^2$ between the 3rd and 4th terms, $5 \cdot 2^3$ between the 4th and 5th terms, and so on. So, the difference between the 31st and 32nd terms will be $5 \cdot 2^{30}$ and $2 + 30 = 32$
9	10 [people]	One morning each member of Gregorio's team at work drank a 12-ounce mixture of coffee with milk. The amounts of coffee and milk varied from mug to mug but were never zero. Gregorio drank one-ninth of the total amount of milk and one-eleventh of the total amount of coffee. How many people are on his team? Let m = the total amount of milk, c = the total amount of coffee, and p = the number of people on his team. Then $(m/9 + c/11)p = m + c \rightarrow (11m + 9c)p = 99m + 99c \rightarrow 11mp + 9cp = 99m + 99c \rightarrow 11mp - 99m = 99c - 9cp \rightarrow 11m(p - 9) = 9c(11 - p)$. Since m and c must both be positive and p must be an integer, then $p - 9$ and $11 - p$ must also both be positive, and this can only happen when $p = 10$.

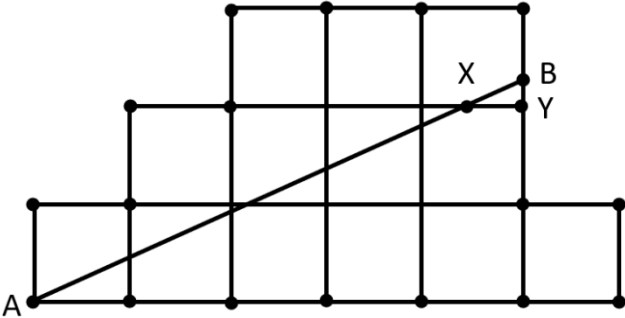
10	4048 [ways]	Each of the 24 students in a math class is assigned a number from 1 to 24 corresponding to a numbered pocket in a classroom phone caddy with 24 pockets. The caddy accidentally falls to the floor and all the phones fall out. The classroom assistant replaces all of the phones to best of his ability, but some of the phones may be in the wrong pocket. In how many ways can exactly 3 of the phones be in the wrong pocket? There are 6 ways that 3 phones can be placed in 3 pockets, one of which is correct. Using letters instead of numbers for pockets and assuming ABC is the correct order, the ways the phones can be placed incorrectly include ACB, BAC, BCA, CAB, and CBA. Only BCA, and CAB have all 3 phones in the wrong pocket, so there are 2 ways for a given set of 3 phones to all be misplaced. There are ${}_{24}C_3 = (24 \cdot 23 \cdot 22)/(3 \cdot 2 \cdot 1) = 4 \cdot 23 \cdot 22 = 92 \cdot 22 = 1840 + 184 = 2024$ sets of 3 phones. The total number of ways that exactly 3 phones can be misplaced is $2 \cdot 2024 = 4048$.
----	-------------	---

“Math is Cool” Championships -- 2025-26

7th/8th Grade

Linda Moore Triple Jump Solutions

	Answer	Solution																
1	18	What is 80% of 22.5? $0.8 \cdot 22.5 = 18$																
2	3	The integers 1 through 10 are written as English words and listed in order alphabetically. As an integer, what is the ninth word in the list? One, two, three, four, five, six, seven, eight, nine, ten → eight, five, four, nine, one, seven, six, ten, three , two																
3	100 [plates]	It takes 3 workers 20 minutes to wash 80 plates. Working at the same rate, how many plates can 5 workers wash in 15 minutes? $3w : 20\text{min} : 80p \rightarrow 1w : 60\text{min} : 80p \rightarrow 5w : 12\text{min} : 80p \rightarrow 5w : 15\text{min} : 100p$																
4	[A =] 27	A cylinder has a radius of 15 millimeters and a height of 0.12 meters. In terms of π and in cubic centimeters , the volume of the prism can be expressed as $A\pi \text{ cm}^3$. What is the value of A? $15 \text{ mm} = 1.5 \text{ cm}$ and $0.12 \text{ m} = 12 \text{ cm}$, so volume = $(1.5)^2\pi \cdot 12 = 2.25\pi \cdot 12 = 27\pi \text{ cm}^3$ and $A = 27$.																
5	24 [years]	In a survey, people were asked about their favorite candy. The number of people who indicated that Skittles is their favorite candy is shown by age group in the histogram below. What is the maximum possible median age in an integer number of years for those who prefer Skittles from the survey? There are $1 + 2 + 4 + 7 + 5 + 3 + 1 = 23$ people who prefer skittles. When the ages are listed in order the median age will be the 12th number in the list. Since the 8th through the 14th numbers in the list are in the age group of 20 – 24, then the largest possible 12th number is 24. Favorite Candy is Skittles <table><thead><tr><th>Age Group (years)</th><th>Number of People</th></tr></thead><tbody><tr><td>5 - 9</td><td>1</td></tr><tr><td>10 - 14</td><td>2</td></tr><tr><td>15 - 19</td><td>4</td></tr><tr><td>20 - 24</td><td>7</td></tr><tr><td>25 - 29</td><td>5</td></tr><tr><td>30 - 34</td><td>3</td></tr><tr><td>35 - 39</td><td>1</td></tr></tbody></table>	Age Group (years)	Number of People	5 - 9	1	10 - 14	2	15 - 19	4	20 - 24	7	25 - 29	5	30 - 34	3	35 - 39	1
Age Group (years)	Number of People																	
5 - 9	1																	
10 - 14	2																	
15 - 19	4																	
20 - 24	7																	
25 - 29	5																	
30 - 34	3																	
35 - 39	1																	

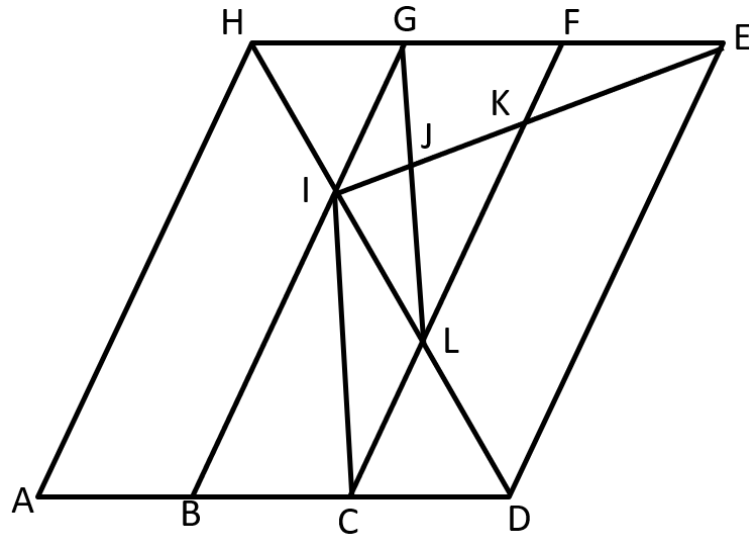
6	9 [quarters]	Emiliano has 22 coins, a mixture of dimes and quarters. If the dimes were quarters and the quarters were dimes, Emiliano would have 60 cents more than he has now. How many quarters does he currently have? $10D + 25Q + 60 = 25D + 10Q \rightarrow 15D - 15Q = 60$ $D + Q = 22 \rightarrow 15D + 15Q = 330$ Adding the two resulting equations gives $30D = 390 \rightarrow D = 13$ and $Q = 9$.
7	[P + Q =] 23	The diagram shows a decagon consisting of 13 unit squares. The portion below $\overline{AB}$ consists of a unit square and a triangle with a base of 5. If $\overline{AB}$ bisects the area of the decagon, then as a reduced common fraction, the area of $\triangle BXY = P/Q$. What is $P + Q$? Half of the area of the decagon is 6.5 units^2. So, the triangle below $\overline{AB}$ has an area of 5.5. If its base is 5, then its height must be $5.5 \cdot 2/5 = 11/5$. This means $BY = 11/5 - 2 = 1/5$. To determine XY solve the proportion $(1/5)/(11/5) = XY/5 \rightarrow 1/11 = XY/5 \rightarrow XY = 5/11$. The area of $\triangle BXY = (1/5)(5/11)/2 = 1/22$ and $1 + 22 = 23$. 
8	[A + B =] 13	The numbers 1 through 9 are marked with points on the number line as shown. Two distinct points are chosen at random from these points. As a reduced common fraction, the expected value of the distance between the two points is A/B. What is $A + B$? Note: the expected value is the average of all possible lengths of segments having any two points as endpoints. $[16(1) + 14(2) + 12(3) + 10(4) + 8(5) + 6(6) + 4(7) + 2(8)] / (16 + 14 + 12 + 10 + 8 + 6 + 4 + 2) = (16 + 28 + 36 + 40 + 40 + 36 + 28 + 16) / 72 = 240 / 72 = 10/3$ and $10 + 3 = 13$. 

9

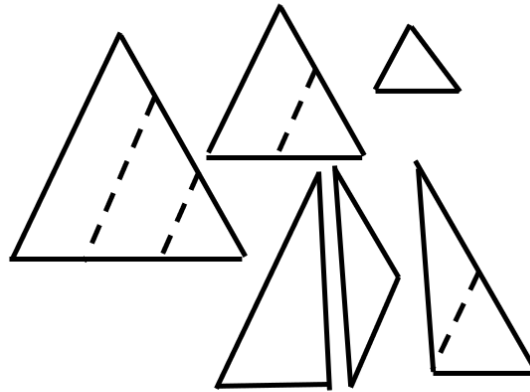
22

[triangles]

How many triangles of any size are in the figure below?

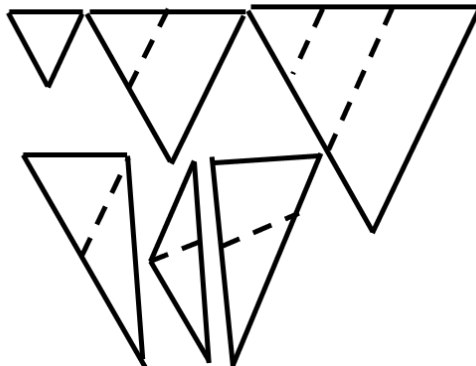


Lower left half - 6



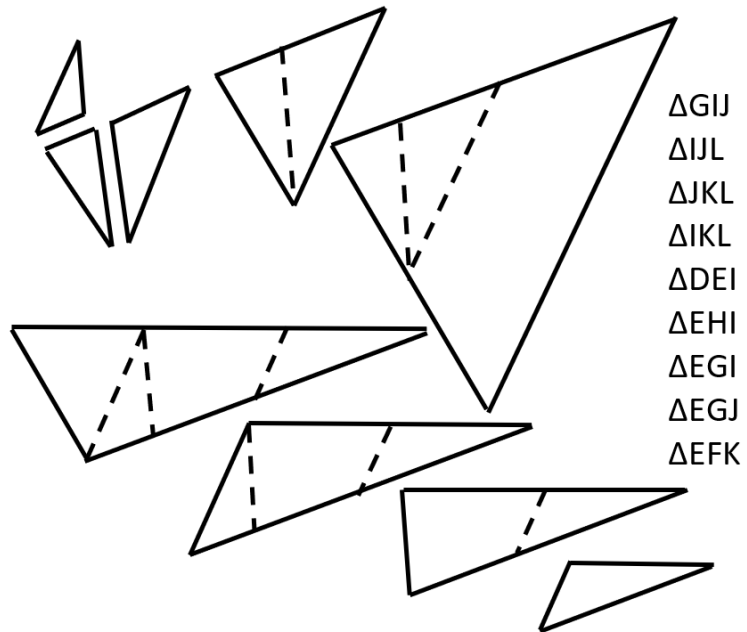
$\triangle ADH$
 $\triangle BDI$
 $\triangle CDL$
 $\triangle BCI$
 $\triangle CIL$
 $\triangle CDI$

Upper right half – 6 that are 180° rotations of the 6 from the lower left half centered on the midpoint of $\overline{IL}$.



$\triangle GHI$
 $\triangle FHL$
 $\triangle DEH$
 $\triangle GHL$
 $\triangle GIL$
 $\triangle FGL$

More in the upper right half - 9



One that spans $\overline{HD} - 1$

ΔCIK



$$6 + 6 + 9 + 1 = 22$$

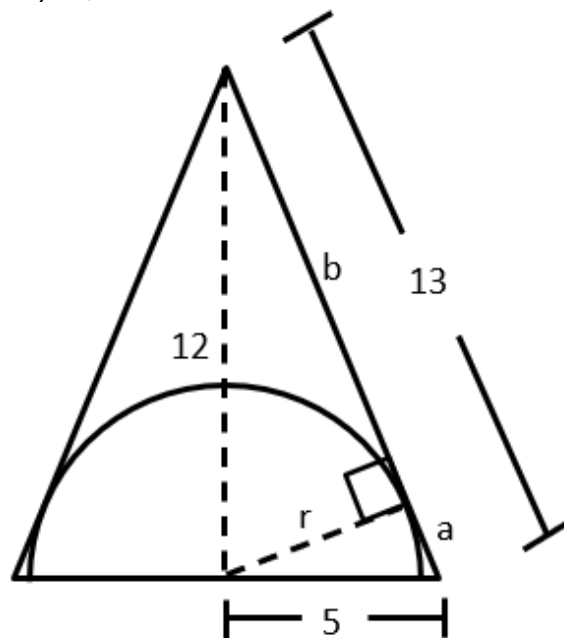
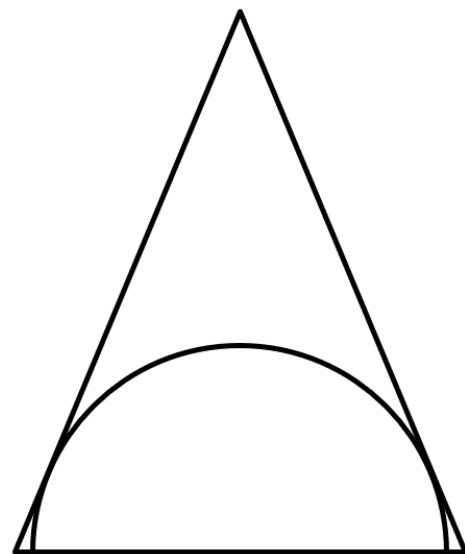
10

[A + B =]
73

A semicircle is inscribed in an isosceles triangle with base 10 cm and height 12 cm so that the diameter of the semicircle is contained in the base of the triangle as shown. As a reduced common fraction, the radius of the semicircle is A/B. What is A + B?

Adding the two auxiliary lines creates two right triangles and you can write a Pythagorean equation for each one, $a^2 + r^2 = 5^2$ and $b^2 + r^2 = 12^2 \rightarrow a = \sqrt{25 - r^2}$ and $b = \sqrt{144 - r^2} \rightarrow$ since $a + b = 13$, then

$\sqrt{25 - r^2} + \sqrt{144 - r^2} = 13 \rightarrow 25 - r^2 + 2 \cdot \sqrt{25 - r^2} \cdot \sqrt{144 - r^2} + 144 - r^2 = 169 \rightarrow 169 - 2r^2 + 2 \cdot \sqrt{25 \cdot 144 - 169r^2 + r^4} = 169 \rightarrow 2 \cdot \sqrt{25 \cdot 144 - 169r^2 + r^4} = 2r^2 \rightarrow \sqrt{25 \cdot 144 - 169r^2 + r^4} = r^2 \rightarrow 25 \cdot 144 - 169r^2 + r^4 = r^4 \rightarrow 25 \cdot 144 = 169r^2 \rightarrow \frac{25 \cdot 144}{169} = r^2 \rightarrow r = \frac{5 \cdot 12}{13} = 60/13$, and $60 + 13 = 73$.



“Math is Cool” Championships -- 2025-26

7th/8th Grade

College Bowl Round #1 Solutions

	Answer	Solution
1	144	What is the sum of the next two numbers in the geometric sequence that begins with three, six, twelve, and twenty-four? $48 + 96 = 144$
2	$[x =] 12$	Solve for X: fifteen X plus sixty equals two hundred and forty $15x + 60 = 240 \rightarrow 15x = 180 \rightarrow x = 12$
3	21260	What is two times thirty plus three times four hundred plus four times five thousand? $2 \cdot 30 + 3 \cdot 400 + 4 \cdot 5000 = 21260$
4	30 [%]	Zeus has ten cards, and each card contains a different one of the first ten prime numbers. If one card is randomly selected, what is the probability in percent that it contains the digit three? The first 10 prime numbers are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Three of them contain a '3'. $3/10 = 30\%$
5	7 [mph]	Shanteel rides her bike for ten minutes at an average rate of six miles per hour and then for five minutes at an average rate of nine miles per hour. In miles per hour, what is her average rate for the whole fifteen minutes? 6 mph for 10 min means she rode 1 mile 9 mph for 5 min means she rode $3/4$ of a mile Together she rode $7/4$ miles in 15 min = 7 mph
6	20 [in^2]	The ratio of the perimeters of two similar rectangles is two to five. If the area of the larger rectangle is one hundred twenty-five square inches, what is the number of square inches in the area of the smaller rectangle? The ratio of the areas equals the square of the ratio of the perimeters, so $(2/5)^2 = 4/25$ is the ratio of the areas. Then $4/25 = x/125 \rightarrow x = 20$.
7	18 [ways]	In how many ways can the letters of the word MATH, spelled M - A - T - H, be arranged such that the first letter is not an M? The total number of ways is $4! = 24$, or 6 for each starting letter (M, A, T, H). $24 - 6 = 18$.
8	84 [= sum]	What is the sum of the even integers between twenty-five and thirty-one? $26 + 28 + 30 = 84$

9	$[A + B =] 17$	$420/56 = 105/14 = 15/2$ and $15 + 2 = 17$
10	2 [= median]	What is the median of the following numbers? One-half, one-fourth, seven-halves, and eight. The four numbers in order are: $1/4$, $1/2$, $7/2$, 8, so the median is $(1/2 + 7/2)/2 = 4/2 = 2$.

“Math is Cool” Championships -- 2025-26

7th/8th Grade

College Bowl Round #2 Solutions

	Answer	Solution
1	9 [candies]	In a bag of thirty-six candies, one-third of the candies are red, one-fourth of the candies are green, and one sixth of the candies are blue. How many candies are not red, green, or blue? $1 - 1/3 - 1/4 - 1/6 = 12/12 - 4/12 - 3/12 - 2/12 = 3/12 = 1/4$ and $36(1/4) = 9$
2	54	What is nine squared minus six squared plus three squared? $9^2 - 6^2 + 3^2 = 81 - 36 + 9 = 54$
3	50 [minutes]	Diego takes eighty minutes to walk four miles. Walking at this same rate, how many minutes would it take him to walk two point five miles? 4 mi in 80 min = 1 mi in 20 min = 0.5 mi in 10 min = 2.5 mi in 50 min
4	17 [chickens]	A field contains only pigs and chickens. In total there are eighty-six legs and sixty eyes among the animals in the field. How many chickens are in the field? $\begin{cases} 4p + 2c = 86 \\ 2p + 2c = 60 \end{cases} \rightarrow 2p = 26 \rightarrow p = 13 \text{ and } c = 17$
5	[A + B =] 7	Two standard six-sided dice are rolled and the numbers showing are added. As a simplified fraction, the probability that the sum of the two numbers is divisible by six is A over B. What is A plus B? Possible sums are 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 and there are 36 total outcomes. There are 5 ways to get 6 and 1 way to get 12 so $P(\text{divisible by 6}) = 6/36 = 1/6$ and $1 + 6 = 7$.
6	41 [= 3 rd term]	In a certain sequence, each term is equal to two minus the quantity three times the previous term. If the first term is five, what is the third term in this sequence? $A_1 = 5$ $A_n = 2 - 3A_{n-1}$ The first 3 terms are: 5, -13, 41
7	11 [times] [e's]	When the number seventy-seven thousand seven hundred seventy-seven is written out in words, how many times does the letter 'e' appear? Seventy seven thousand seven hundred seventy seven

8	10 [toes]	One foot is equivalent to twelve inches. Let two point four inches be called a "TOE". How many TOES are in two feet? $2 \text{ ft} = 24 \text{ inches}$ and $24/2.4 = 10$
9	27 [= A]	The circumference of a circle with a radius of thirteen point five centimeters is $A\pi$ centimeters, where A is a two-digit integer. What is A ? $r = 13.5$, $d = 27$, $C = 27\pi$, so $A = 27$
10	10200	What is the smallest product of two distinct, positive three-digit composite numbers? $100 \cdot 102 = 10200$

“Math is Cool” Championships -- 2025-26

7th/8th Grade

College Bowl Round #3 Solutions

	Answer	Solution
1	6384	What is the product of seventy-six times eighty-four? $76 \cdot 84 = (80 - 4)(80 + 4) = 6400 - 16 = 6384$
2	52 [Fridays]	If January 1, 2026, was a Thursday, how many total Fridays will there be in 2026? $364/7 = 52$, so after January 1, there will be 52 complete weeks, meaning every day of the week will occur 52 times, except Thursday will occur 53 times.
3	64 [= mean]	What is the mean of the numbers forty, seventy, and eighty-two? $(40 + 70 + 82)/3 = 192/3 = 64$
4	15 [seconds]	Jenna runs ninety meters in eighteen seconds. In seconds, how long does it take Kira to run ninety meters at an average rate that is twenty percent faster than Jenna's average rate? Jenna's rate = $90/18 = 5$ m/s Kira's rate = $1.2(5) = 6$ m/s and $90/6 = 15$ s.
5	[x =] 39	Solve for X: X over six minus three over five equals two X over fifteen plus seven over ten $x/6 - 3/5 = 2x/15 + 7/10 \rightarrow 5x - 18 = 4x + 21 \rightarrow x = 39$
6	[A + B =] 17	The least common multiple of two positive integers, A and B, is seventy. What is the smallest possible value of A plus B? The prime factorization of 70 is $2 \cdot 5 \cdot 7$. To keep the sum of the two integers A and B as small as possible, there must be no duplications of the three prime factors distributed among them, i.e. $A = 2 \cdot 5 = 10$ and $B = 7$, or $A = 2 \cdot 7 = 14$ and $B = 5$, or $A = 5 \cdot 7 = 35$ and $B = 2$. Of these possibilities $10 + 7 = 17$ is the smallest.
7	[A + B =] 21	Five fair coins are tossed and each lands showing either heads or tails. As a simplified fraction, the probability that exactly four of the five coins show the same result is A over B. What is A plus B? $P(4H, 1T) = 5 \cdot (1/2)^5 = 5/32$ and $P(4T, 1H) = 5 \cdot (1/2)^5 = 5/32$, and $5/32 + 5/32 = 5/16$ and $5 + 16 = 21$.
8	18 [cm]	The perimeter of a regular heptagon is one hundred and twenty-six centimeters. In centimeters, what is the length of one side of the heptagon? $126/7 = 18$

9	$[A - B =] 12$	Two distinct positive integers, A and B, are both less than fifty and greater than thirty-six. What is the greatest possible value of A minus B? $49 - 37 = 12$
10	680 [= 10 th term]	The first three terms of an arithmetic sequence are fifty, one hundred twenty, and one hundred ninety. What is the tenth term in the sequence? $50 + 9(70) = 680$

“Math is Cool” Championships -- 2025-26

7th/8th Grade

College Bowl Round #4 Solutions

	Answer	Solution
1	777	What is the largest three-digit integer in which all three digits are prime numbers? Single-digit primes are 2, 3, 5, and 7, so the largest 3-digit integer in which all three digits are prime is 777.
2	1296 [in ²]	How many square inches are in a square yard? $36 \cdot 36 = 1296$
3	38 [= N] [= smallest value of N]	When a positive integer N is divided by ten the remainder is eight. When N is divided by nine, the remainder is two. What is the smallest value of N? Check numbers that end in 8, since they will have a remainder of 8 when divided by 10: 8, 18, 28, 38, ... The first of these numbers that has a remainder of 2 when divided by 9 is 38.
4	7 [terms]	A sequence is defined by adding five to the numerator and four to the denominator of a given fraction to get the next term. The last term in the sequence is the first integer term in the sequence. If the first term is one-seventh, how many terms are in the sequence? $1/7, 6/11, 11/15, 16/19, 21/23, 26/27, 31/31 = 1$, so 7 terms total.
5	20 [rooms]	A Roomba vacuum cleaner vacuums one room in three-tenths of an hour. At this rate, how many rooms will it vacuum in six hours?
6	[A + B =] 9	A fair coin is flipped four times. As a simplified fraction, the probability that no two consecutive flipw have the same result is A over B. What is A plus B? There are $2^4 = 16$ total outcomes. Only two of them have no consecutive flips the same: HTHT or THTH. $2/16 = 1/8$.
7	22 [drills]	Naomi has a weekly set of drills she uses to improve her tennis skills. After finishing three drills she has completed twelve percent of her weekly set. How many drills does she have left to complete for the week? $3 \text{ drills} = 12\% \rightarrow 1 \text{ drill} = 4\% \rightarrow 25 \text{ drills} = 100\%$, so she has $25 - 3 = 22$ drills left.

8	12 [ways]	How many distinct ways can four beads be arranged in a straight line if two of them are identical yellow beads, one is green, and one is blue? YYGB, YYBG, BYYG, GYYB, BGYY, GBYY, YGYB, YBYG, GYBY, BYGY, YBGY, YGBY makes 12 ways.
9	71 ^[°]	The measures of angles A and B in triangle ABC are thirty-two and seventy-seven degrees. What is the number of degrees in the measure of angle C? $180 - 77 - 32 = 71$
10	[x =] 7	Solve for X: seven X plus eleven minus four X equals two to the fifth $7x + 11 - 4x = 2^5 \rightarrow 3x + 11 = 32 \rightarrow 3x = 21 \rightarrow x = 7$

“Math is Cool” Championships -- 2025-26

7th/8th Grade

College Bowl Round #5 Solutions

	Answer	Solution
1	2970	What is the product of fifty-five times fifty-four? $55 \cdot 54 = 2700 + 270 = 2970$
2	48 [cats]	A certain animal shelter houses only cats and dogs. If there are a total of one hundred and twelve animals at the shelter and four-sevenths of them are dogs, how many cats are in the shelter? $\frac{3}{7} \cdot 112 = 48$
3	22 [= 4 th number]	The mean of three numbers is eighteen. When a fourth number is added to the set, the mean becomes nineteen. What is the fourth number? $3 \cdot 18 = 54$ and $(54 + N)/4 = 19 \rightarrow 54 + N = 76 \rightarrow N = 22$
4	23	What is two thousand eight hundred and twenty-nine divided by one hundred and twenty-three? $2829/123 = 943/41 = 23$
5	8 [%]	As a percent, what is the common difference in the arithmetic sequence whose first three terms are one over twenty-five, three over twenty-five, and one-fifth? In the sequence $1/25, 3/25, 1/5 = 5/25$, the common difference is $2/25 = 8\%$.
6	[A + B =] 181	The letters in the word COTTON, spelled C-O-T-T-O-N, are used to form a six-letter arrangement. As a simplified fraction, the probability that the letters are in alphabetical order is A over B. What is A plus B? Only 1 arrangement is in alphabetical order and there are $6!/(2! \cdot 2!) = 180$ total arrangements, so the probability is $1/180$ and $1 + 180 = 181$.
7	45 [minutes]	Safran can paint ten boards in thirty minutes. Cayenne can paint eight boards in twenty minutes. In minutes, how long would it take the two working together to paint thirty-three boards? Safran paints $10/30 = 1/3$ board per minute and Cayenne paints $8/20 = 2/5$ board per minute. Together they paint $1/3 + 2/5 = 11/15$ boards per minute and $11/15 \cdot \# \text{ of minutes} = 33 \text{ boards} \rightarrow \# \text{ of minutes} = 45$.

8	4 [times]	The numbers from one to twenty-one are written on a piece of paper. How many times does the digit two appear? A 2 appears in the numbers 2, 12, 20, and 21, for a total of 4 times.
9	[x =] 46	Solve for X: two X minus seventy-two equals the square root of four hundred. $2x - 72 = \sqrt{400} \rightarrow 2x - 72 = 20 \rightarrow 2x = 92 \rightarrow x = 46$
10	945 [cm ³]	A rectangular prism has a length of ten centimeters, a width of seven centimeters, and a height of thirteen point five centimeters. In cubic centimeters, what is the volume of the prism? $L \times W \times H \rightarrow 10 \cdot 7 \cdot 13.5 = 135 \cdot 7 = 945.$

“Math is Cool” Championships -- 2025-26

7th/8th Grade

College Bowl Round #6 Solutions

	Answer	Solution
1	32 [pages]	Beyoncé reads twelve pages in nine minutes. At this rate, how many pages will she read in twenty-four minutes? 12 p in 9 min $\rightarrow$ 4 p in 3 min $\rightarrow$ 32 p in 24 min
2	42 [= gcf]	What is the greatest common factor of one hundred twenty-six and two hundred ten? $126 = 2 \cdot 3^2 \cdot 7$ and $210 = 2 \cdot 3 \cdot 5 \cdot 7$, so the GCF = $2 \cdot 3 \cdot 7 = 42$
3	[B - C =] 13	Let A, B, and C represent three distinct positive integers, all less than sixteen. If A is greater than B and C is less than B, what is the greatest value of B minus C? (A, B, C) = (15, 14, 1) results in the largest value of $B - C = 14 - 1 = 13$.
4	24 [= number]	A number is divided by negative six, then the quotient is increased by six. The result is two. What was the original number? $n/-6 + 6 = 2$ $n/-6 = -4$ $n = 24$
5	224 [inches]	A rectangle has an area of three square feet, and its length is twenty-seven times its width. What is the number of inches in the perimeter of the rectangle? $w(27w) = 3 \rightarrow 27w^2 = 3 \rightarrow w^2 = 1/9 \rightarrow w = 1/3 \text{ ft} = 4 \text{ inches}$ and $l = 27(4) = 108$ and perimeter = $2(4) + 2(108) = 224$.
6	40 [%]	Let A be fifteen percent of eighty and let B be twenty-five percent of one hundred and twenty. As a percent, what is A divided by B? $A = 0.15(80) = 12$ and $B = 0.25(120) = 30$ and $A/B = 12/30 = 2/5 = 40\%$.
7	31 [= mode]	A data set consists of six positive integers and has a unique mode. If the mean is twelve, what is the greatest possible value of the mode? The numbers must add up to 72. Since the mode is unique and in order to maximize it, the other four numbers must be distinct and as small as possible: 1, 2, 3, 4, 31, 31.
8	50 [zinks]	Two plinks are equal to six winks. Three winks are equal to ten zinks. How many zinks are equal to five plinks? $2p = 6w$ and $3w = 10z \rightarrow 2p = 20z \rightarrow 5p = 50z$

9	273 [= sum]	What is the sum of the integers three through twenty-three? $21(3 + 23)/2 = 21(13) = 273$
10	3000 [cm ³]	There are one hundred centimeters in a meter. How many cubic centimeters are equivalent to three-thousandths of a cubic meter? $3/1000 \cdot 100 \cdot 100 \cdot 100 = 3/1000 \cdot 1000000 = 3000$